

Article

Against the Odds: Hybrid Zones between Mangrove Killifish Species with Different Mating Systems

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Citation: Berbel-Filho, W.M.;

Tatarenkov, A.; Pacheco, G.;

Espírito-Santo, H.M.V.; Lira, M.G.;

Garcia de Leaniz, C.; Avise, J.C.; Lima, S.M.Q.; Rodríguez-López, C.M.;

Consuegra, S. Against the Odds:

Hybrid Zones between Mangrove

Killifish Species with Different

Mating Systems. *Genes* **2021**, *12*, 1486.

[https://doi.org/10.3390/](https://doi.org/10.3390/genes12101486)

[genes12101486](https://doi.org/10.3390/genes12101486)

Academic Editor: Malgorzata Pilot

Received: 16 August 2021

Accepted: 22 September 2021

Published: 24 September 2021

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Abstract: Different mating systems are expected to affect the extent and direction of hybridization. Due to the different levels of sexual conflict, the weak inbreeder/strong outbreeder (WISO) hypothesis predicts that gametes from self-incompatible (SI) species should outcompete gametes from self-compatible (SC) ones. However, other factors such as timing of selfing and unilateral incompatibilities may also play a role on the direction of hybridization. In addition, differential mating opportunities provided by different mating systems are also expected to affect the direction of introgression in hybrid zones involving outcrossers and selfers. Here, we explored these hypotheses with a unique case of recent hybridization between two mangrove killifish species with different mating systems, *Kryptolebias ocellatus* (obligately outcrossing) and *K. hermaphroditus* (predominantly self-fertilizing) in two hybrid zones in southeast Brazil. Hybridization rates were relatively high (~20%), representing the first example of natural hybridization between species with different mating systems in vertebrates. All F1 individuals were sired by the selfing species. Backcrossing was small, but mostly asymmetrical with the SI parental species, suggesting pattern commonly observed in plant hybrid zones with different mating systems. Our findings shed light on how contrasting mating systems may affect the direction and extent of gene flow between sympatric species, ultimately affecting the evolution and maintenance of hybrid zones.

Keywords: asymmetric introgression; *Kryptolebias*; mixed mating; reproductive isolation; self-fertilization

1. Introduction

Hybridization is a major source of evolutionary innovation, with important implications for phenotypic diversification, adaptation and ultimately speciation [1,2]. Prezygotic barriers, such as spatial, temporal, and behavioral differences play an important role in regulating the extent of hybridization amongst sympatric species in nature [3]. In addition, differences in mating systems (defined as the proportion of self-fertilization versus outcrossing [4]) can largely influence the extent of hybridization, as well as the direction of gene